

Thein Mwae Than Tha

M: 8168523 | E: 1009361@mymail.sutd.edu.sg | LI: <https://www.linkedin.com/in/thein-mwae-than-tha-7b82232ab/?l=https://theinportfolio.netlify.app/> | Portfolio:

EDUCATION

Singapore University of Technology and Design (SUTD)

Engineering Product Design (EPD)

- Relevant Skills: Embedded Systems • Product Engineering • Rapid Prototyping • CAD • Hardware Integration • IoT Systems

Singapore

Sep 24 - Present

Singapore Polytechnic

Diploma in Computer Engineering

- Relevant Skills: Python, C++, Arduino, Raspberry Pi, IoT, MySQL, OOP

Singapore

Mar 21 - Apr 24

TECHNICAL SKILLS

Programming Languages: Python | C++ | Arduino | SQL | Embedded C

Hardware & Embedded: Raspberry Pi | Sensors Integration | Microcontrollers | IoT Systems

Engineering & CAD: Fusion 360 | Rapid Prototyping | Hardware Prototyping | System Integration

Software & Tools: MySQL | OOP | Cloud Data Upload | Circuit Development

Technical fields: Product Engineering | Embedded Systems | Hardware-Software Integration | Smart Systems

PROJECTS

Smart Assistive Standing Device

Product Engineering / Embedded Systems / Mechatronics

- Engineered a portable assistive sit-to-stand system integrating linear actuator mechanisms, analog control circuitry, and mechanical support structures
- Designed and prototyped scissor-lift linkage architecture using CAD and rapid fabrication workflows for load-assisted lifting applications
- Assisted in electronics integration involving PWM-based actuator control, sensor interfacing, and motor driver circuitry
- Conducted iterative prototype refinement through hardware validation, mechanical redesign, and subsystem-level troubleshooting
- Developed functional engineering prototype combining mechanical, electrical, and control-system integration principles

Smart Vertical Farming Monitoring System

IoT / Embedded Systems / Hardware Prototyping

- Engineered an IoT-based environmental monitoring platform for vertical farming applications using integrated sensor architecture
- Developed a real-time hardware monitoring prototype capable of automated environmental sensing and cloud-based data transmission
- Integrated sensor monitoring, image capture, and remote data upload into a unified embedded monitoring workflow
- Iterated hardware prototype configurations to improve monitoring reliability and real-time system responsiveness

WORK EXPERIENCE

Singapore Polytechnic

Inhouse Internship, Engineering Intern

- Developed and tested IoT-enabled monitoring prototypes for smart farming applications
- Supported sensor integration, hardware implementation, and real-time monitoring workflows

Mar 23 - Aug 23

Singapore, Singapore

Project Osiris

Start Up Company, Hardware Engineer

- Created modular hardware architecture integrating sensors, microcontroller systems, and power distribution modules within a startup engineering team.
- Made circuit schematics and implemented embedded hardware integration for functional prototype deployment .
- Fabricated and tested prototype subsystems for system-level integration and hardware validation.
- Demonstrated integrated engineering prototype at Singapore Management University showcase event.

Oct 23 - Jan 24

AWARDS & ACTIVITIES

- 1st Prize — National IIOT Competition (2022/2023)
- WorldSkills Convention Participant Overseas Engineering Study Trip — George Town, Malaysia
- Overseas Engineering Study Trip — George Town, Malaysia